Qiuyi Luo

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Myself

I obtained a PhD degree from the Shanghai Astronomical Observatory, University of Chinese Academy of Sciences. My research interests focus on understanding the formation of multiple systems in our Galaxy, exploring both local and global environmental effects. I am an advance user of radio observation with interferometers (ALMA) and single-dish telescopes (e.g., NRO-45m, JCMT) and infrared observation.

Work

- **Center for Astrophysics | Harvard & Smithsonian** *03/2026–*
  Radio Department Boston, USA
- Visiting Researcher

- **The University of Tokyo** *09/2025-03/2026*
  Institute of Astronomy / Department of Astronomy Tokyo, Japan
- Project Researcher

Education

- **Changchun Normal University** *09/2014-06/2018*
  Physical Department Changchun, China
- B.S., Physics

- **University of Chinese Academy of Science** *09/2019-06/2025*
  Shanghai Astronomical Observatory Beijing, China
  - Ph.D., Astronomy

- **National Astronomical Observatory of Japan** *10/2023-04/2025*
  ALMA project Mitaka, Japan
  - Visiting Student, Radio Astronomy

Press Releases

- **Astronomers Discover Forming Quadruple-star System: More Intimate, More Complex**. Press Release by Chinese Academy of Sciences(Luo et al. 2023) 08/2023

- **JCMT Astronomers find that denser and more turbulent environments tend to form multiple stars**. Press Release by East Asian Observatory(Luo et al. 2022) 06/2022

Honors and Awards

- **Outstanding Student Award** *2023*
  Joint of Bank of China - Chinese Academy of Science
- **The First Prize Scholarship** *2023*
  University of Chinese Academy of Sciences
- **Merit Student** *2022*
  University of Chinese Academy of Sciences

Observing Program

PI Programs:

◦ VLA, 33 hrs, **Identifying High-Mass YSOs in Multiple Systems Embedded in Massive Protoclusters**

◦ SMA, 8 hrs, **Gas kinematics of the quadruple protostellar system**

◦ FAST, 3.4 hrs, **Unveil scenario of cloud formation in multi-shell system G24**

Co-PI programs

◦ ALMA, Cycle 11, **Uncovering the Role of the Magnetic Field in Streamers**

◦ ALMA, Cycle 11, **Digging into the Newly Discovered Shock-associated Molecular Inventories near Extremely Young Protostellar Outflows**

◦ ALMA, Cycle 10, **Searching for Complex Organic Molecules in Orion Cold Cores**

◦ ALMA, Cycle 9, **Searching for complex organic molecules in Orion cold cores**

◦ ALMA, Cycle 8, **Searching for complex organic molecules in Orion cold cores**

◦ FAST, 16hrs, **Multi-band spectroscopic surveys of the lambda Orionis complex**

◦ FAST, 10hrs, Zeeman Observations of HI Narrow Self-Absorption towards mini-starbrust W43-main

◦ FAST, 20.5hrs, **FAST Observations of HI towards a CO-harboring HVC – the G165**

Talks at Conferences

◦ **Oct. 2025 The Low-mass multiple star system formation in our Milky Way**, Perspectives in Star Formation — Senior Insights, Young Visions, Shanghai, China

◦ **Oct. 2025 The Low-mass multiple systems formation in High-mass cluster-forming Regions**, Fifth Chile–Japan Academic Forum, Kyoto, Japan

◦ **Aug. 2025 The Formation of Low-mass Multiple Systems in High-mass Cluster-forming Regions**, Multiplicity in Young Stars 2025, online

◦ **Aug. 2025 The Formation of Low-mass Multiple Systems in High-mass Cluster-forming Regions**, Multiplicity in Young Stars 2025, online

◦ **Aug. 2024 The Multiplicity of Low-mass Protostars in High-mass Star-forming Regions**, ALMA-ATOMS/QUARKS Workshop 2024, online

◦ **Jul. 2024 DIHCA: Mapping Low-mass Multiplicity in High-mass Star-forming Regions**, EA ALMA Science and Data Analysis Workshop 2024, Seoul, South Korea

◦ **Nov. 2023 The Formation of Multiple Stellar Systems in the Orion GMC**, Stellar Origins: NAOJ & Tokyo Tech Star Formation Workshop, Tokyo, Japan

◦ **Aug. 2023 A forming quadruple system with intricate ‘ribbons’ and outflows**, From molecular cloud cores to star formation 2023, Nanjing, China

◦ **Apr. 2023 High-order system G206 in Orion (poster)**, Protostars and Planets VII, Kyoto, Japan

◦ **Jun 2022 The Origin of Multiple Stellar System in the Orion Molecular Cloud Complex**, The SHAO-XIAMEN Astronomy Symposium 2023, Xiamen, China

◦ **Jun. 2022 ALMASOP: The Formation of binary/multiple systems in Orion molecular cloud complex**, JCMT Users Meeting, online

Main Publications

[J.1] Qiuyi Luo et al. (2026). **Digging into the Interior of Hot Cores with ALMA: The Formation of Low-mass Multiple Systems in High-mass Cluster-forming Regions.** *The Astrophysical Journal*, 999:192, DOI: 10.3847/1538-4357/ae371f

[J.2] Qiuyi Luo et al. (2023). **ALMA Survey of Orion Planck Galactic Cold Clumps (ALMASOP): A Forming Quadruple System with Continuum “Ribbons” and Intricate Outflows**. *The Astrophysical Journal Letters*, Vol. 952, Issue 2, L2, DOI: 10.3847/2041-8213/acdddf

[J.3] Qiuyi Luo et al. (2022). **ALMA Survey of Orion Planck Galactic Cold Clumps (ALMASOP): How Do Dense Core Properties Affect the Multiplicity of Protostars?**. *The Astrophysical Journal Letters*, 931:158, DOI: 10.3847/1538-4357/ac66d9